NCLEX study guide

Addison's Disease	Cushing's Disease
-think hyperthyroidism sx	-think hypothyroidism sx
Hyposecretion of glucocorticoids -not enough aldosterone = lose water (think diuretics..some block aldosterone)	Hypersecretion of glucocorticoids -too much aldosterone = water retention
Hypovolemia	Hypervolemia
Hot	Cold (**C**ushing's = **C**old)
↑K ↑Ca ↓ Na (lose H2O, lose Na)	**↓K ↓Ca ↑Na** (gain H2O, gain Na)
Hypoglycemia (↑ insulin production)	Hyperglycemia (↓ insulin production) + ketoacidosis
Wet skin	Dry skin (hyper = dryer) hyperglycemia = dry skin
Lethargy, fatigue, muscle weakness	Generalized muscle wasting, weakness
Hypotension (↓Na)	Hypertension (↑ Na)
Weight loss	Weight gain / Slow healing
Decreased blood volume + shock Hyperkalemia = meta acidosis + arrhythmias	Moon face, buffalo hump, obesity (trunk), thin skin, reddish-purple striae
TREATMENT: hormone replacement	TREATMENT: hypophysectomy, adrenalectomy
Addisonian crisis -medical emergency -critical deficiency of glucocorticoids -generally follows acute stress, sepsis, trauma, surgery, or omission of steroid therapy **s/sx:** severe abd pain, sudden profound weakness, hyperpyrexia followed by hypothermia, coma, renal failure	Osteoporosis (excess cortisol = ↑Ca reabsorption from bones)

Hyperthyroidism	Hypothyroidism
- fast	- slow
-↑ metabolism	↓ metabolism
-sympathetic NS sx	-parasympathetic NS sx
-nervousness, irritable, excitable, tachycardia, perspiration, flushed face, exophthalmus, increased appetite, limp hair, wt loss, HTN	-extreme fatigue, dry skin, coarse hair, numbness and tingling of fingers, alopecia, wt gain
-heat intolerance	-cold intolerance
Iodine uptake ↑	Iodine uptake ↓
Graves disease	**Myxedema coma**

Thyroid storm -tachycardia -delirium -coma	
-pt with hyperthyroidism is typically nervous and has insomnia. -don't place in same room with another pt with hyperthyroidism because too much stimuli. -place in private room.	

Removing tubes and things

Chest tubes	Have pt perform valsalva maneuver, or take and hold deep breath (have seen both in nclex books)
NG tube	Have pt take and hold a deep breath
PICC line	Have pt perform valsalva maneuver
TPN line	Valsalva maneuver

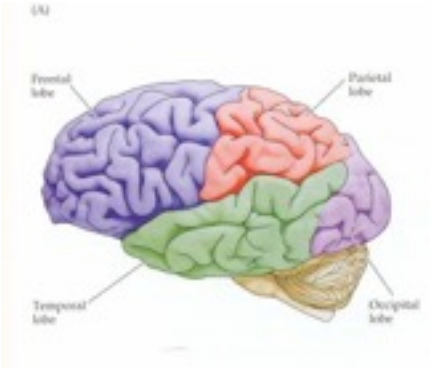
Hyperglycemia hyperosmolar nonketotic syndrome (HHNKS)	DKA
-occurs in people with DM-2	-occurs in peoplee with DM-1
-glucose > 800 ml/dL	-glucose > 300-800 mg/dL
-gradual onset of sx	-sudden onset of sx
No ketosis/acidosis	Ketosis/acidosis / FRUITY breath odor
Polyuria, polydipsia, dehydration, mental status changes, wt loss, weakness, headache ↓BP	Polyuria, dehydration, wt loss, dieresis ↓BP / Tachycardia
Tx = fluid replacement, correct electrolyte imbalance, give insulin /Exercise	Tx = give vasopressin IV Fluids / Reg. Insulin
Skin warm/ dry and dry mucus memb / high Temp	Kussmaul respirations (rapid and deep)
	-expected outcome = ↑ responsiveness

Sympathetic NS (fight/flight)	Parasympathetic NS
-anticholinergic drugs **-vasoconstriction**	**-B blockers** **-vasodilation**
Tachycardia	Bradycardia
Dilated pupils	Constricted pupils
Inhibits digestion -constipation	Stimulates digestion -diarrhea
Inhibits nasal secretions	Stimulates nasal secretions

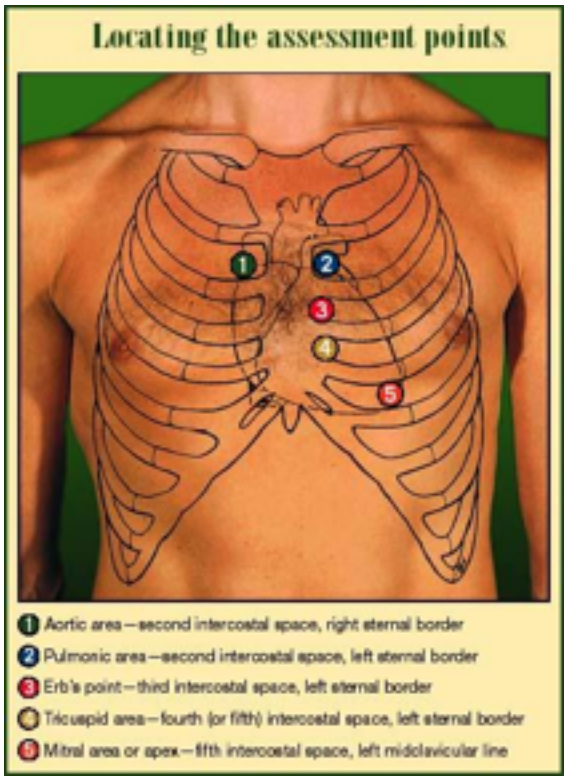
Inhibits saliva production	
Inhibits liver, kidneys, gallbladder	
Stimulates sweating	Stimulates liver, kidneys, gallbladder
Lungs dilate	Constricts lungs
Increases muscle strength	

Breath sounds

Tracheal breath sounds	-very loud and high pitched -can be heard over the trachea -I = E
Vesicular breath sounds	-soft and low pitched -normal breath sound -heard over most of the lungs -I > E
Bronchial breath sounds	-very loud and high pitched -heard over the manubrium -if heard elsewhere, may indicate consolidation -E > I
Bronchovesicular breath sounds	-intermediate intensity and pitch -best heard in the first and second ICS (anterior chest) and between scapula (posterior chest) -I = E



<u>Parietal lobe</u> = primary center for sensation
<u>Temporal lobe</u> = auditory reception areas
<u>Frontal lobe</u> = involved with personality, behavior, emotions, intellectual function, if injured may have difficulty comprehending what is said
<u>Occipital lobe</u> = primary visual receptor center



MASLOW'S

Lab values

K	3.5 – 5.1 (no pee, no K)
Na	135 – 145
Mg	1.6 – 2.6
Cl	98 – 107
Ca	8.6 – 10
Wbc	4500 – 11000
Serum osmolality	285 – 295 ↑ = dehydration ↓ = overhydration
Hgb	14-16.5
Hct	42-52% (3 x Hgb)
Amylase	25-151 ↑ with pancreatitis (acute 5x nml, chronic 3x nml) -aids in digestion
Albumin	3.4 – 5
Fe	65-175
ALT/AST	5-60/5-43 (liver function tests)
Uric acid	4.5 – 8 (men) 2.5 – 6.2 (women) ↑ with gout

Lipase	10 – 140 ↑ in liver dz (Lipase = Liver)
↓Ca	+ Trousseau's sign = arm + Chvostek's sign = cheek
CK enzymes	MB = cardiac muscle BB = brain MM = skeletal
Troponins	-better indicator of detecting MI than CK's
Cl^- Cystic Fibrosis	>60
IOP/ICP	10-20
Urine specific gravity	1.010 – 1.025 > 1.030 dehydration
Mantoux skin test	>5mm with AIDS/HIV >10mm normal
Sedimentation rate (ESR)	-rate increases with more inflammation Men: 0-15mm/hr Women: 0-20mm/hr
HbA1c	-reflection of how well blood glucose levels have been controlled for the past 3-4 months -diabetics with good control = **7% or less** -increases = not good control -hyperglycemia causes it to ↑ -High Triglycerides causes false + (high levels) -Kidney disease causes false – (low levels)

1cc	1ml
1cc	15gtt
30cc	1oz
5cc	1tsp
15cc	1Tbs
1mg	1000mcg
60mg	1 grain
bid	2x a day
tid	3x a day
qid	4x a day
qod	every other day
ac	With each meal
hs	At bedtime

	BP	HR
↑ ICP	↑	↓
Shock	↓	↑
Bleeding	↓	↑
Cushing's Triad (also widening pulse P)	↑	↓
Autonomic dysreflexia	↑	↓
Air embolus	↓	↑

Eye stuff

Mydriatic eye drops	Big word = big pupils Dilates pupils
Miotic eye drops	Little word = little pupils Constricts pupils
Glaucoma	-silent thief of vision -optic nerve damage -causes irreversible blindness -blurred vision, halos, loss of peripheral vision -risk factor = <u>cardiovascular dz</u> -treat with meds to decrease IOP (B-blockers) and miotics (increase outflow of aq humor) -African Americans are at an increased risk at any age -nursing goal: prevent further deterioration
Cataracts	-lens opacity or cloudiness -painless, blurry vision -surroundings are dimmer -diplopia
Macular degeneration	-dry = nonexudative (slow) -wet = exudative (fast) -drusen = tiny yellow spots

Arterial leg ulcers	Venous leg ulcers
-small	-large
-circular	-irregular
-deep	-superficial
	-granulation tissue
	-highly exudative
Pain = intermittent claudication (pain caused by activity)	Pain = aching, heaviness

Nephritis	Nephrotic Syndrome (Nephrosis)
Think "I" (thin kid)	Think "O" (round kid)
Periorbital edema, facial edema	↑ edema (ascites), periorbital edema
High BP	Low BP
Anorexia	Lethargy, pallor, anorexia
↓UO	↓UO
Hematuria	Swollen abd, labia, scrotum
Pallor, irritability, lethargy	
Proteinuria	Massive proteinuria
↑BUN, creatitine, ASO titer (pt reports strep infection before)	
Treatment: antibiotics, antiHTN	Treatment: steroids

Important Drugs

-olol = B-blocker (↓HR, ↓BP)
-pine = Ca channel blocker (↓HR, ↓BP)
-pril = ACE inhibitor (↓BP) vasodilate
-sartan (similar to ACE inhib) – for pt allergic to ACE inhib.

Digoxin	-**Toxicity** = nausea/anorexia (early signs), green halos, ↓UO -monitor K and Mg (low levels and increase toxicity) -therapeutic level < 2 (0.8 – 1.5) -If given with lasix, monitor K! -pt with hypothyroidism is more sensitive to dig (↓K) -Ca can make toxicity worse (↓Ca = ↓K) -usual dose = 0.25mg/day -↓ workload of heart and ↑myocardial function -↑ intracellular Ca -assess apical pulse before admin
Theophylline	-for asthma or COPD -don't give with food/drinks that contain caffeine

MgSO4 *for eclampsia *tocolytic	-therapeutic level = 4-7.5 Toxicity -flushing -↓RR, DTR, BP, UO -pulmonary edema
Drugs that stimulate B1 and B2	B1 = 1 heart (increase HR) B2 = 2 lungs (dilate lungs)
Clozapine	-antipsychotic -risk for agranulocytosis -give pt anticholinergic to help this (Benztropine – Cogentin) – also give this with Thorazine
Tetracycline	-take on empty stomach -dairy can bind with it and prevent absorption -no Ca, Mg, Al, Fe (prevent absorption)
Li toxicity	-therapeutic level = 0.5 – 1.2 Toxicity = N/V, muscle weakness, severe diarrhea, tinnitus, blurred vision
Lidocaine toxicity	-drowsiness and CNS disturbances
Lasix toxicity	-renal failure (↓OU), blood dyscrasias, hearing loss
Methergine (prevent postpart bleed)	↑ ctx (**m**ethergine = **m**ore) (increase contractions, prevent postpart bleed, subinvolution)
Terbutaline	↓ ctx (stop contractions/ preterm labor)
Bethanechol	-cholinergic med -10-50mg 3-4x/day -tx for urinary retention
Fosamax	-take on empty stomach!
Morphine	Toxicity = pinpoint pupils, ↓RR
Dilantin	-causes urine to change colors -only give with NS
Diuretics	-some block aldosterone (Aldactone) = K sparing -aldosterone retains H2O and Na, loses K
Infusion rate too fast?	Hypotension
Clomipramine (Anafranil)	"Ana" is depressed because of her OCD -tricyclic antidepressant -can also be used for OCD
Clonidine	-HTN and opiate w/d
Coumadin	Monitor PT Antidote = vitamin K
Heparin	Monitor aPTT Antidote = protamine sulfate

Mental Health

Schizophrenic patients	Remember **SDS** to remember major needs: **S** = structure: because they tend to have too little in their lives **D** = diversion: to distract them from disturbing thoughts **S** = stress reduction: to minimize the severity of the disorder
Paranoid pt	don't encourage , don't support their coping mechanisms Ask direct questions: "do you hear voices?"
Panic attacks	-sympathetic NS sx -HTN, ↑HR, ↑alertness, SOB, trembling
Alcohol w/d sx -DT	-give Librium
Types of crisis	**Situational** (external source, unanticipated) -divorce -loss of job -death of loved one -abortion -severe physical, mental illness **Maturational** (occurs at a developmental stage) -marriage -birth of child -retirement **Adventitious** (crisis of disaster) -not part of everyday life -flood, fire -9/11 -rape
Manic pt	-have them reorganize something
Akathisia	Restlessness, pacing, inability to sit still
Dystonia	-tonic contractures of muscles in the neck, mouth, tongue
Parkinsonism	-muscle rigidity, shuffling gait, stooped posture, flat-faced affect, tremors, drooling
What factor has the most influence on the outcome of the a crisis situation?	-previous coping skills
In extreme stress don't do what?	-teach, education…learning is limited at this point
Serotonin syndrome	-diarrhea -irritability, restlessness -tremors, delirium -fever -tachycardia, HTN -apnea -death
Double bind communication	-pt says "I love you" but has a grimace on face
RESTRAIN	Answer is always least restrictive intervention (this include meds)

Cranial Nerves

I	Olfactory	Smell
II	Optic	Central/peripheral vision
III	Oculomotor	Pupil constriction
IV	Trochlear	Have pt follow tip of finger
V	Trigeminal	Jaw strength
VI	Abducens	6 cardinal movements of eyes
VII	Facial	Facial symmetry
VIII	Acoustic	Ears –hearing
IX	Glossopharnygeal	Taste, uvula midline, etc
X	Vagus	Taste, uvula midline, etc.
XI	Accessory	Neck, shoulder
XII	Hypoglossal	Midline tongue

Precautions

Standard	-uniform level of caution that should be used in <u>all</u> patients -primary goal = prevent transmission of nosocomial infection -hand hygiene -gloves -misc barriers (mask, eye protection, face shield, gown)
Contact	-in addition to standard -used for organisms that are easily spread by skin-to-skin contact, or by contact with items in pt's environment -may place pt in private room -masks are not needed, doors do not need to be closed **<u>Examples</u>** -antibiotic-resistant organisms -enteric infections with low infectious dose -c-diff -GI, respiratory, skin, wound infections or colonization with multidrug-resistant bacteria -RSV -highly infectious skin infections: diphtheria, herpes, impetigo, pressure ulcers, scabies, shingles -conjunctivitis -ebola

Airborne	-in addition to standard -for pt with serious illnesses transmitted by airborne droplet nuclei **<u>Examples</u>** -measles -varicella (and disseminated zoster) -TB
Droplet	-in addition to standard <u>Examples</u> -flu -pertussis -adenovirus -mumps / rubella

Food

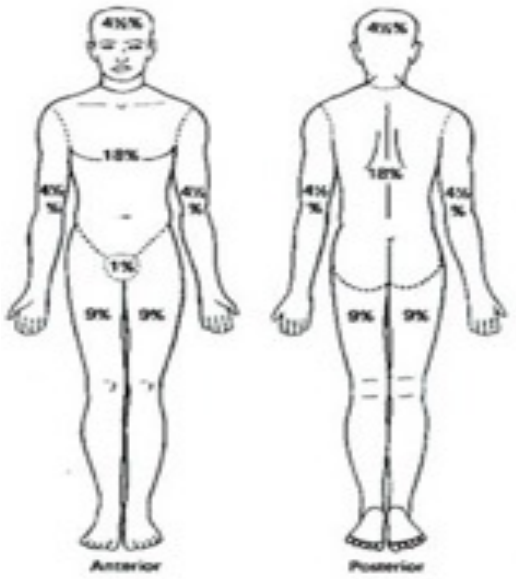
Tyramine (Korsakoff Psychosis= tyramine deficiency)	-avoid with **<u>MAOIs</u>**, migraines -figs, avocados, bananas, papaya, raisins -aged cheese, yeast, yogurt, sour cream -soybeans, beer, red wine -beef, liver, sausage, bologna, deli meat -chocolate
Purine	-avoid with **<u>gout</u>** -fish, sardines -liver, beef, chicken, sausage, organ meats
Gluten	-avoid with Celiac's disease
Vitamin K (antidote for Coumadin)	-broccoli, cabbage, turnips -fish, liver -coffee, tea (caffeine)
Vitamin B12 (thiamine)	-found in animals, nuts, whole grain cereals -pt with cirrhosis needs a diet high in B12
Calcium	-eggs -green leafy veggies
Potassium	-potatoes -bananas -spinach -raisins -dates -oranges -dried apricots
Iron	-can give with Vitamin C (tomato juice, OJ) -clams -liver, beef, shrimp -turkey -cereal -pasta
Folic acid	-liver -papaya -legumes, vegs, spinach -nuts, bran, cereal -fruit, yeast, asparagus
Acid ash diet	-avoid milk = makes urine alkaline

Vitamin D toxicity	-GI upset and metallic taste -HA, weakness, renal insufficiency, renal calculi, HTN, arrhythmias, muscle pain, conjunctivitis
Crohn's diet	-Low fat -Low residue (no popcorn) -High protein
Calcium	-take 1 hour after meals with full glass of water
ACE inhibitors	-take 1hour before meals
Rhogam	Mom (-) Baby (+)

Burns

Superficial partial thickness	-first degree -sunburn -epidermis -red, blanches with pressure -possible blisters
Deep partial thickness	-second degree -scald -epidermis, upper dermis, part of deeper dermis -blistered, mottled red base -weeping, edema
Full thickness	-third degree -flame, chemicals, electrical current -epidermis, entire dermis, muscle/bone -dry, pale white -leathery, fat exposed, edema

*burn pt at risk for ↑K



Misc Med/Surg diseases

Guillain-Barre Syndrome	-↑ weakness (ascending) -neuro problem = acute infection of cranial and peripheral nerves -pt c/o respiratory infection or GI infection in past med hx -immune system overreacts and destroys myelin sheath -major concern = problems breathing s/sx = paresthesis, lower extremity weakness, gradual progressive weakness, possible resp failure, cardiac probs, high protein in CSF
Myasthenia Gravis	-eye problems -sedatives make sx worse -neuro prob -weakness and fatigue -have pt do things in am -defect in transmission of nerve impulses -give meds before meals s/sx = weakness, fatigue, difficulty chewing, dysphagia, weak/hoarse voice, resp failure, ptosis, diplopia, decreased breath sounds -everything pretty much slows down and gets weak **<u>Tensilon test</u>** = used to dx -if pt shows improvement after tx = dx
Hirschspring's dz	-mega-colon -results in mechanical obstruction b/c of inadequate motility s/sx = failure to gain weight, abd distention, vomiting, <u>ribbon-like</u> and foul smelling stools (not with newborns I think…), constipation alternating with diarrhea
B-thalassemia	-too much Fe -autosomal recessive disorder -decreased production of 1 of the globin chains in the synthesis of Hgb -chelation drug therapy (gets rid of Fe)

Hip replacement	-avoid extreme external, internal rotation -avoid adduction -no side-lying on operative side -maintain abduction with pt in supine position or on non-operative side -do not cross legs -place pillow b/w legs to maintain abduction
Wernicke-Korsakoff syndrome	-neuro disorder -acute encephalopathy -chronic psychosis -caused by deficiency in Vit B / Tyramine deficiency
Multiple myeloma	-↑Ca caused by bone destruction is the primary concern -encourage fluids (dilutes Ca)
Pancreatitis	-do not give morphine! (irritates pancreas) -pain is severe and unrelenting in epigastric area and radiates to back -observe for ↓UO, ↑HR -diet = ↓fat, ↓protein, ↑carbs, K supplements -typically rest GI by making pt NPO but give lots of IV fluids (+) Turner's sign = bruiselike discoloration in flank (+) Cullen's sign = bluish hemorrhage around umbilicus
Dumping syndrome	-limit fluids with meals -early sx = sweating and pallor -5-30 min after eating -also vertigo, tachycardia, desire to lie down
Fat embolus	-tachypnea -tachycardia -dyspnea
Air embolus	-chest pain -dyspnea -lightheadedness -nausea -dizziness -hypoxia -anxiety -↑HR ↓BP
Liver biopsy	-position pt on right side to help decrease risk of hemorrhage -don't let pt cough = can cause bleeding
5th dz	-not contagious after rash
Peptic ulcer dz	Primary sx of perforation = board-like abd and shoulder pain (blood) Coffee ground emesis = slower internal bleeding
Asthma	-diminished wheezing in a child with asthma indicates possible worsening of asthma
Posturing	De<u>cere</u>brate = <u>cere</u>bellum problem De<u>cort</u>icate = <u>cort</u>ex problem -Decorticate is more favorable than decerebrate (thought it was other way around?)

Pheochromocytoma	-produces catecholamines (epi) -tumor – adrenal medulla -headache, diaphoresis, palpitations, HTN, tremor, hyperglycemia -don't palpate abd = can cause more catecholamines (cells) to be released and cause severe HTN
Parathyroid	-monitor Ca and P (Ca↑ P↓) **<u>Hyperparathyroidism (↑Ca, ↓P)</u>** -increased sleeping -<u>increased urination</u> -weakness -bone pain -irritability **<u>Hypoparathyroidism (↓Ca, ↑P)</u>** -<u>increased urinary frequency</u>
SIADH	-decreased UO = fluid overload, HTN, ↑HR -water intoxication -↓Na (dilutional hyponatremia) -too much antidiuretic hormone (vasopressin) -treatment is effective if: ↑UO, ↓wt, ↓urine osmolality
Diabetes insipidus	-deficiency of ADH = ↓ ability of kidneys to concentrate urine -give vasopressin s/sx = polyuria, polydipsia, fatigue, dilute urine, intense thirst, dehydration, wt loss, HA, tachycardia, ↑Na in urine
Tumor lysis syndrome	-emergency of electrolyte imbalance -potential renal failure
Pulmonary Embolism	-blood tinged sputum -chest pain -cough -cyanosis -distended neck veins -dyspnea -↓BP, ↑HR -wheezes -tachypnea
Renal failure	**<u>Pre</u>** -poor perfusion to kidneys **<u>Intra</u>** -damage to kidneys, nephrotoxic injury from contrast, antibiotics, corticosteroids **<u>Post</u>** -obstruction of urinary collecting system
Autonomic dysreflexia	-hyperreflexia -spinal cord injury T5 and above (I think) -overactivity of autonomic NS -kinked cath can cause it, constipation or full bladder (Incr ICP) -pounding HA, HTN, sweating, bradycardia, restlessness

COPD	-use a high-flow venture mask to deliver O controlled, specified amount of O **s/sx**: dyspnea on exertion, barrel chest, clubbed fingers and toes

Cardiovascular

Complications of mitral stenosis	-thromboembolism -rheumatic fever (common complication of CHF) -endocarditis -pulmonary HTN -pulmonary edema
Hemolytic transfusion rxn	-headache -tachycardia -HTN and Hypotension -apprehension, sense of impending doom -fever, chills -DIC -**low back pain,** chest pain
Autologous transfusion rxn	- s/s of infestion (greatest risk)
Inotropic and Chronotropic Drugs	**Inotropics** -affect force of muscle ctx (-) inotropic effects = ↑ myocardial contractile force (+) inotropic effects = ↓ myocardial contractile force (b-blockers) **Chronotropics** -affects HR (-) chronotropic effects = ↓ HR (parasym NS, acetylcholine) (+) chronotropic effects = ↑ HR (sym NS), epi, atropine) Digoxin (+) inotropic (-) chronotropic Drugs for HTN (-) inotropic (-) chronotropic
Diagnostic tests	1. Troponins are more specific 2. CK-MB
Coronary arteriogram	-femoral artery is used – keep pt on bedrest with HOB slightly elevated for several hours -↑HR in recovery may be a sign of hemorrhage (common complication)
Cardiac tamponade	-fluid around heart -pt may c/o heavy / fullness around heart

First priority of care for pt with cardiovascular problem	↓ cardiac workload ↑ myocardial oxygenation

L-sided Heart Failure	R-sided Heart Failure	HF in children
Left = Lung	Jugular vein distention	- gallop rhythm
Dyspnea	Edema	
Tachypnea	Wt. gain	
Gallop rhythm: S3, S4	Ascites	
Fine crackles	Hepatomegaly	
Wheezing, rhonchi	Tachycardia	
Tachycardia	Fatigue	
	Oliguria (fluid retention)	
*acute pulmonary edema	*mitral stenosis	

Medications and Insulin

Meds that ↑ Insulin requirements	Meds that ↓ Insulin requirements
Glucocorticoids (cushings = hyperglycemia)	Sulfonylrureas
Li	Quinidine
Rifampin (TB)	Quinine (malaria)
Progestins (oral contraceptives)	ACE inhibitors
Nicotine	Naproxen
Phenytoin	Indomethacin (gout, RA, OA)
Ca-channel blockers	Salicylates
Clonidine	B-blockers
Morphine	
Heparin	

*exercise = insulin needs increase
*baby born to diabetic mom is at risk for hypoglycemia (give extra feedings of formula)

Diabetes - Insulin

	Onset	Peak	Common types	Misc.
Rapid Midmorn-trembling/wkness	15min	1-2h	Aspart (novalog) Lispro (humalog)	Clear, sliding scale, no IV, pump, can mix with I, L

Short Early evening – wkness, fatigue	30-60min	2-4h	Regular	Only kind that can be given IV Clear Can mix with I, L
Intermediate Early evening – wknes, fatigue	1-2h	4-8h (4-12h?)	NPH (Humalin R, Novalin R) Lente	Cloudy Can mix with R, S
Long	2-4h	8-14h	Humalin U	Cloudy Can mix with R, S
Very Long-actig	1-2h	None (ongoing)	Glargine (Lantus)	Clear Never mix with others! Usually given at bedtime

*RN – draw up "R"egular first and "**N"PH** second
*Oral hypoglycemics = stimulate pancreas to produce more insulin or increase sensitivity to insulin already there, **only** for DM-2
*DM-2 and insulin needs during surgery, stress, infection = ↑ need for insulin
*Reduce your insulin needs during exercise (exercise lowers blood glucose)
*Glucagon = prevents hypoglycemia, produced by the pancreas, action is opposite of insulin

Delegation

RN	LPN	NA
Orient pt to floor	Dressing changes	Vital signs
Assess foods high in K pt who can't have K	Check neurovascular status (I think)	Ambulate pt
Education, teaching, etc.	Patient in traction -check skin -observe sites for infection -clean insertion sites	Remind pt to use incentive spirometer
Blood products	Administer MDI meds	Gather supplies
	Check O2 sat via pulse ox	Hygiene
	Give meds	
	Enemas	

*delegating to a float nurse = look for the implied experience and skills that are applicable from one clinical area to another when assigning patients (general teaching, assessment)
*don't delegate to LPN: teaching, assessing, unstable pt

IV Solutions

Isotonic	
D5W	-don't use during fluid resuscitation -used mainly to supply water and correct ↑ serum osmolality
0.9% NaCl (NS)	-used with blood transfusions -used with Dilantin -used to replace Na losses -burn injuries -doesn't supply calories -not for: HF, pulmonary edema, renal impairment, Na retention

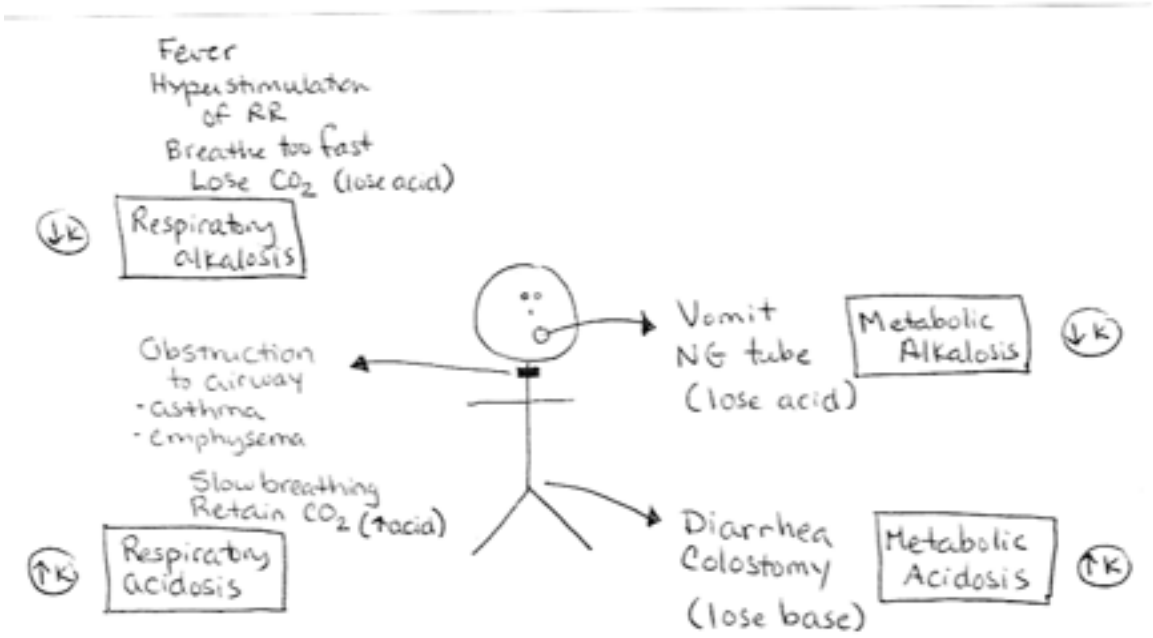
LR	-corrects dehydration, Na depletion -replace GI losses
Hypotonic	
0.45% NaCl	-dehydration
Hypertonic	
D5W 0.9% NaCl	

ACID / BASE BALANCE

ROME

Respiratory opposite
Metabolic equal

↓ pH	↑ CO2	Respiratory acidosis
↑ pH	↓ CO2	Respiratory alkalosis
↓ pH	↓ HCO3	Metabolic acidosis
↑ pH	↑ HCO3	Metabolic alkalosis



MISC

Fire in patients room? PACE / RACE P = get patient out / R = Rescue patients A = activate fire alarm, rescue other patients C = close door to confine fire E = extinguish fire

MISC:
Presence of glucose in nasal discharge = fluid is CSF
<u>Catecholamines</u> -dopamine, epi -released during times of stress
Thyroidectomy – monitor Ca and P
<u>Chemo treatment</u> -↑ uric acid levels in blood d/t massive cell destruction
<u>Calmette-guerin vacc</u> = vacc for TB -mantoux test will always be positive
CO_2 in blood = vasodilation
Allergy to bananas/kiwis = allergy to latex
Acute pain sx = ↑BP, ↑HR, ↑RR, ↑perspiration, ↓body T, ↑dilated pupils (wide eyed with fear)
If a question asks you to select a goal for a pt, make sure the answer you pick is an actual goal! 1. maintain O_2 2. keep HOB elevated to promote proper ventilation = no, this is an intervention
Allergy to eggs = no flu shot
Dx test to confirm TB = sputum culture
Infiltration = cool to touch, swelling, tenderness, decreased rate, blanching of skin **Phlebitis** = inflammation, redness, heat, swelling, tenderness
HTN-crisis - give phentolamine: vasodilates
Best area to check a dark-skinned patient for: Petechiae = oral mucosa, conjuntivae Cyanosis = palms/soles of hands and feet Jaundice = sclera

Strategies

Look for umbrella answer – if all the answers are correct, does one contain the others?
Which one is not like the other?
Look for opposites, look for similar answers to find the one that isn't the same (rapid pulse, tachycardia…)
ABCs
Like dz can room with like dz . (Clean pt with Clean pt / Dirty with Dirty)
Assess before you implement! - Unless no further assessment is needed
Safety 1st
Maslow's - IMPORTANT

Avoid key words – always, never, only - Throw these out
Look for words like "pt suddenly developed chills…" the "suddenly" means new and serious! Priority!

Priority

An unconscious pt with L sided tracheal shift from midline or a pt clutching her chest and c/o severe chest pain?	The pt with L sided tracheal shift = airway
Priority interventions if pt has pulmonary edema	1. admin O2 2. foley cath (to monitor I/O since giving diuretics) 3. Lasix 4. Morphine - ↓ work of breathing and anxiety
Priority of actions if pt with DM-1 who received NPH and regular insulin 2h ago c/o hunger, weakness, shakiness	1. check blood glucose level 2. give pt 1/2c fruit juice 3. take vital signs 4. retest blood glucose 5. give pt small snack of carbs/protein 6. document
TRIAGE	T = trauma R = respiratory I = ICP and mental status A = an infection G = GI , upper E = elimination, lower

Types of play

Parallel	-toddlers -side by side -rarely interact
Associative	-preschoolers -all engaged I similar activity, but little organization
Cooperative	-school-age -organized and goal-directed
Therapeutic	-technique used to help understand a child's feelings
Play Therapy	- Allow the child the express themselves easier

Blood

Complications of a blood transfusion	**1. <u>Transfusion rxns</u>** -weak pulse, fever, brady/tachycardia, hypotension, oliguria **2. <u>Circulatory overload</u>** -cough, chest pain, wheezing, HA, HTN, ↑HR, distended neck veins **3. <u>Septicemia</u>** -chills, fever, vomiting, shock, hypotension **4. <u>Fluid overload</u>** **5. <u>Dz transmission</u>** -Hep B, for example **6. <u>Hypocalcemia</u>** -citrate in transfused blood binds with Ca and is excreted -hypereflexia, paresthesia, tetany, muscle cramps, +Trousseau's sign, +Chvostek's sign 7. **<u>Hyperkalemia</u>** -stored blood liberates K
Pt with severe blood loss requires rapid transfusion. What device is used during blood transfusions to decrease risk of cardiac dysrhythmias?	-blood warming device -rapid transfusions of cool blood puts pt at risk for cardiac dysrhythmias
IV solution that can only be run with blood transfusions	0.9% NaCl
How long do you have to admin blood products once they are picked up from blood bank?	15-30min
Reason to delay a blood transfusion	Fever – hold and notify dr
Special about blood transfusion tubing	Has a built-in filter
Pt is receiving platelets…what might the pt exhibit to show he is benefiting from the transfusion?	-decrease of bleeding from puncture sites and gums, etc.
What would you use to evaluate effectiveness of fresh frozen plasma	Coag studies (PT, PTT)

Pediatric/Women's Health Stuff

Newborn Apgars (1min, 5min)
Heartrate (< or > 100)
Respiratory effort
Muscle tone
Reflex irritability
Skin color

0-3
4-6
7-10

Fontanels
Anterior = diamond (closes 18mon)
Posterior = triangle (closes 6mon)

Pregnancy T to watch out for = greater than 100.4

Postpartum changes
-↑ Hct, hunger, UO
-↓ blood vol, progesterone production

Fundus
-descends 1cm/day below umbilicus postpartum

After breaking water = assess FHR!

5 P's of labor
Passage
Passenger
Power
Psyche
Position

Plantar creases
Preterm = red creases, not very distinguished
Term =2/3 of foot
Postterm = entire sole

Layers of placenta = amnion (inner), chorion (outer)

Shiny = baby
Serum glucose 40-60

Preliminary signs of labor
-Braxton hicks ctx
-dropping (lightening)
-nesting
-lose 1-2lb
-bloody show

Stages of Labor
Stage 1 – ctx to complete dilation)
-Latent (0-3cm)
-Active (4-7cm)
-Transition (8-10cm)
Stage 2 – delivery of baby
Stage 3 – delivery of placenta
Stage 4 – post partum

7 Cardinal Movements (edfieee)
Engagement
Descent
Flexion
Internal rotation
Extension
External rotation
Expulsion

Bracton Hicks ctx = 23-27w

Quickening = 18-22w (13-25w)

Fetal heartbeat = 18-22w

Lochia
Rubra - red
Serosa - pink
Alba - yellow

Sources of folic acid
-liver, papaya, legumes, vegs, spinach, breakfast cereal

Decelerations (fetal heart)
Early: head compression (normal)

Late: uteroplacento insufficiency
Variable: cord compression
Interventions:
-turn pt
-increase primary IV
-d/c pitocin
-amnioinfusion
-elevate legs
-call dr

Why give neonate vit K?
-neonate lacks intestinal flora to make it
-at risk for bleeding disorders
-necessary for blood coag
-vit K stimulates liver to produce clotting factors

Fetal monitoring
Normal = 110-160
Usual tone of uterine activity = 10-12
Bradycardia < 110 @ least 10min
Tachycardia > 160 @ least 10min

Treatment of maternal hypotension
STOP
S = stop pitocin
T = turn pt on L side
O = administer O
P = push IV fluids if hypovolemic

Placenta Previa
-when the placenta implants in the lower uterine segment where it encroaches on the internal cervical os
-one of the most common causes of bleeding during 2
-painless bright red bleeding after 20th
suddenly)

*risk factors: multigestation, multi pregnancies

Low: placenta implants in lower uterine segment
Partial: placenta partially occludes cervical os
Total: placenta totally occludes cervical os

Causes of postpartum hemorrhage
-uterine atony
-risk increases with increased # of pg
-lacerations of birth canal
-formation of a hematoma

Risk:
-large infant
-time of labor (fast/slow)
-retained placenta
-# of pg
-trauma

Chadwick's sign: blue
Goodell's sign: cervical softening
Hegar's sign: softening of lower uterine segment

5 year old – should be able to memorize their phone #

Primary intervention with a pediatric burn pt = remove blisters

Centration = tendency to center attention on 1 feature of something and be unable to see its other qualities

Teach a pt with sickle cell how to prevent sickling: maintain hydration, promote oxygenation, avoid strenuous activity

PKU = autosomal recessive

Most important newborn reflex = **MORO** (determines neuro dev)

Placenta previa vs placenta abrupto? **PAIN!**

GP_{TPAL}

Placenta previa
-after 20weeks
-painless
-no vag exam
-↓BP
-low/partial/total

Placenta abruption
-after 20weeks
-painful
-d/t trauma/cocaine/HTN
-↑BP
-Kleihauer-bette test/FDP
-mild/mod/strong

Hyperemesis gravidarum
-d/t ↑ HCG
-severe N/V
-+ ketones

Hydatidiform mole
-exaggerated sx of pg
-complete (grapes, empty egg, no fetus/placenta/etc parts)
-partial (2 sperm/1egg, abnl fetus/ placenta, baby dies in utero)

Spontaneous abortion
-bleeding after 20w
-threatened
-inevitable
-complete
-incomplete
-missed

Ectopic pregnancy
-implantation of fertilized ovum outside uterine cavity
-no vag bleeding
-common site: fallopian tube
-methotrexate: inhibits growth
-↓HCG, cullen's sign

HTN disorders
Chronic HTN
-present before/during/after pg
-before 20w
Gestational HTN (PIH)
-after 20w
-no proteinuria
-resolves after pg
->140/90
Preeclampsia
-HTN and proteinuria
-low platelets
->140/90
-after 20w
-prefer vag delivery
Chronic HTN w/ preeclampsia
-new onset proteinuria
-sudden ↑BP
Severe preeclampsia
-clonus
-need 1 of 9 sx:
->160/110, pulmonary edema
->5g/24h proteinuria
-abnl LFTs, oliguria, IUGR
-RUQ/epigastric pain
-thrombocytopenia
-visual/cerebral disturbances
Eclampsia
-seizures
-$MgSO_4$
-proteinuria

HELLP
-complication of preeclamsia
-hemolysis, elevated liver enzymes, low platelets

Diabetes
1st trimester
needs ↓
2nd trimester
begin to ↑
3rd trimester
and taper off at 37w

BPP
-determines fetal demise
-"fetal apgar score"
-use u/s
-fetal muscle tone
-fetal activity
-fetal breathing motions
-look at fluid pocket
-NST

Incompetent cervix
-premature cervical dilation
-painless
-around 20w
-d/t AMA, trauma
-cerclage

Maternal death
1. Hemorrhage
2. Infection
3. HTN

NST
-assesses fetal well-being
-FHR accels in response to fetal movement

Menorrhagia
-heavier than normal menstrual bleeding

Dysmenorrhea
-painful menstrual bleeding

When are fetal movements palpable on abd? **28 weeks!**

Transition phase of labor
-breathing should be pant/blow to resist urge to push

Pediatric stuff

Burn pt – priority intervention is to remove blisters

Myelomeningocele pt – priority intervention is to measure head circumference

Promote motor dev in preterm infant – prevent grasping

Do NOT give pregnant pt diuretics!
-Can cause ctx!

Diuretics, NSAIDs, drugs affecting electrolytes

Lasix	-loop diuretic -↓K
HCTZ	-thiazide diuretic -↓K -exacerbates gout
Aspirin	-NSAID -↓ inflammation -Reye's syndrome in kids
Iron sulfate	-Fe deficiency anemia -take with vitamin C -best taken b/w meals -no antacids -make cause black stools
Fosamax	-↑Ca -take in morning on empty stomach
KCl	-never give IV push -use in pt with hypokalemia
Kayexalate	-use in pt with hyperkalemia
Tylenol	-not anti-inflammatory -pain relief, fever
B12	-↓K -for vit B12 deficiency
Aldactone	-K sparing diuretic
Mannitol	-osmotic diuretic -↓IOP/ICP -renal failure

CNS drugs

Dilantin	-antiepileptic -only give with NS -s/e = blurred vision, diplopia
Sinemet (levodopa/ carbidopa)	-parkinson's dz
Imitrex	-HA
Aricept	-alzheimer's
Nubain	-opioid
Narcan	-opioid antagonist
Morphine	-opioid -toxicity = pinpoint pupils, decreased RR, increased ICP
Prozac	-antidepressant, SSRI

Elavil	-antidepressant, tricyclic -anti-chol s/e

Autonomic drugs

Bethanechol	-parasympathetic -for urinary retention -cholinergic
Epi	-sympathetic -inhalation = fastest route
Atropine	-sympathetic -anti-cholinergic
Dopamine	-sympathetic